Tutorial 7: Sensitivity Analysis

MAT3007 Optimization

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Review: Duality

Primal-Dual Transformation:

Primal	minimize	maximize	Dual
constraints	$\geq b_i$ $\leq b_i$ $= b_i$	≥ 0 ≤ 0 free	variables
variables	≥ 0 ≤ 0 free	$\leq c_j$ $\geq c_j$ $= c_j$	constraints

Weak Duality Theorem (for the standard form LP): If x is feasible to the primal and y is feasible to the dual, then $b^T y \leq c^T x$.
Strong Duality Theorem: If a linear program has a finite optimal solution, so does its dual, and the optimal values of the primal and dual are equal.

	Finite Optimum	Unbounded	Infeasible
Finite Optimum	$\surd$		
Unbounded			$\surd$
Infeasible		$\surd$	$\surd$

Review: Duality

Complementarity Condition: Let x and y be feasible points of the primal and dual problem. Then x and y are optimal iff.

$$y_i \cdot (a_i^T x - b_i) = 0, \quad \forall i \tag{1}$$

$$x_j \cdot (A_j^T y - c_j) = 0, \quad \forall j \tag{2}$$

Remark:

- For a feasible x, y, complementarity condition is the necessary sufficient condition of optimality.
- Complementarity condition can help find optimal dual solution given primal solution quickly.

Exercises 1

Let A be a symmetric square matrix. Consider the linear programming problem:

$$\begin{aligned} \min \quad & c^T x \\ & Ax \geq c \\ & x \geq 0 \end{aligned}$$

Prove that if x^* satisfies $Ax^* = c, x \geq 0$, then $x*$ is the optimal solution.

Exercise 1: Solution

Primal:

$$\begin{aligned} \min \quad & c^T x \\ & Ax \geq c \\ & x \geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \max \quad & c^T y \\ & A^T y \leq c \\ & y \geq 0 \end{aligned}$$

Therefore, if x^* satisfies $Ax^* = c, x \geq 0$, then x^* is a feasible solution to both problems, and the objective value of two problems are equal. By strong duality, the solution is optimal.

Exercise 2

1.Show that exactly one of the following two systems has a solution:
(a) $Ax > 0$ (b) $A^T y = 0, y \underset{\neq}{\geq} 0$. $\quad (a \underset{\neq}{\geq} b$ means $a \geq b$ and $a \neq b)$

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How to handle the strict inequalities?

$Ax \geq 1$ has no solution $\iff$ $Ax > 0$ has no solution.

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How to handle the strict inequalities?
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Solution
We construct the following primal dual
(P)

$$\begin{aligned} \min \quad & 0^T x \\ s.t. \quad & Ax \geq 1 \end{aligned}$$

(D)

$$\begin{aligned} \max \quad & 1^T y \\ s.t. \quad & A^T y = 0 \\ & y \geq 0 \end{aligned}$$

If (P) is feasible:
(a) holds, and the optimal value of (D) $= 0$ (Strong Duality) $\implies$ $y = 0$.
So (a) and $\neg$(b).
else (P) is infeasible: (D) is unbounded $\implies$ $\neg$ (a) and (b)

Sensitivity Analysis

Sensitivity Analysis: We want to analyze how do the optimal solution and the optimal value change when the input changes.

Primal

$$\begin{aligned} \min \quad & c^T x \\ s.t. \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & b^T y \\ s.t. \quad & A^T y \leq c \end{aligned}$$

We mainly consider the change of b and c.

Local Sensitivity: (V here is the objective value)

- if dual has a unique optimal solution, then $\nabla_b V(b) = y^*$.
- if primal has a unique optimal solution, then $\nabla_c V(c) = x^*$.

When the change is small, the change of objective value can be directly computed by $\Delta V = \Delta b^T y^*; \Delta V = \Delta c^T x^*$

Sensitivity Analysis

Global Sensitivity Analysis: What range of changes belong to small change?

Analysis of Δb:

As long as the optimal basis doesn't change, the local sensitivity result holds. We check

$$\tilde{x_B} = A_B^{-1}(b + \Delta b) = x_B + A_B^{-1}\Delta b$$

$$V(b + \Delta b) = c_B^T \tilde{x_B} = V(b) + c_B^T A_B^{-1}\Delta b = V(b) + y^{*T}\Delta b$$

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$$V(b + \Delta b) = c_B^T \tilde{x_B} = V(b) + c_B^T A_B^{-1}\Delta b = V(b) + y^{*T}\Delta b$$

Since b doesn't affect the reduced cost, we only need to check the feasibility of the basic solution for original optimal basis:

$$\tilde{x_B} = x_B + A_B^{-1}\Delta b \geq 0$$

When analyzing a particular b_i, we set $\Delta b = \lambda e_i$ (e_i is a vector with 1 at position i and 0 otherwise). For local sensitivity to hold, we have

$$x_B + \lambda A_B^{-1} e_i \geq 0$$

Sensitivity Analysis

Analysis of Δc:

As long as the original optimal primal solution doesn't change, the local sensitivity result holds. We check

$$V(c + \Delta c) = c_B^T x_B^* + \Delta c^T x_B^*$$

Sensitivity Analysis

Analysis of Δc:

As long as the original optimal primal solution doesn't change, the local sensitivity result holds. We check

$$V(c + \Delta c) = c_B^T x_B^* + \Delta c^T x_B^*$$

Since c doesn't affect the feasibility of original solution, we only need to check the optimality of it. To do this, we check the reduced cost (of non-basic part)

$$\tilde{c}_N^T - \tilde{c}_B^T A_B^{-1} A_N \geq 0$$

We consider $\Delta c = \lambda e_j$.

(1) $j \in B$

$$c_N^T - (c_B^T + \lambda e_j^T) A_B^{-1} A_N = \overbrace{c_N^T - c_B^T A_B^{-1} A_N}^{r_N^T} - \lambda e_j^T A_B^{-1} A_N \geq 0$$
$$\iff r_N^T - \lambda e_j^T A_B^{-1} A_N \geq 0$$

(2) $j \in N$

$$c_N^T + \lambda e_j - c_B^T A_B^{-1} A_N \geq 0$$
$$\iff r_N^T + \lambda e_j \geq 0.$$

Exercise 3: Continue

An insurance company is introducing three products: special risk insurance, mortgage insurance, and long-term care insurance. The expected profit is \$500 per unit on special risk insurance, \$250 per unit on mortgage insurance and \$600 per unit on long term care insurance. The work requirements are as follows: The management team wants to establish sales quotas for each product to maximize the total expected profit.

Department	Working hours per unit			Working hours available
	Special risk	Mortgage	Long-term care	
Underwriting	2	1	1	240
Administration	3	1	2	150
Claims	1	2	4	180

(1). Formulate this problem as a linear optimization problem. Specify the decision variables, objective function, and constraints

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$$
\begin{aligned}
\max \quad & 500x_1 + 250x_2 + 600x_3 \\
s.t. \quad & 2x_1 + x_2 + x_3 \leq 240 \\
& 3x_1 + x_2 + 2x_3 \leq 150 \\
& x_1 + 2x_2 + 4x_3 \leq 180 \\
& x_1, x_2, x_3 \geq 0
\end{aligned}
$$

Exercise 3: Continue

(2). After solving the problem, the final simplex tableau (for the standard form) is given as below (the variables are in the natural order as in the description of the problem). Find the range of working hours available for claims to keep the current basis optimal.

B	0	50	0	0	140	80	35400
4	0	0.5	0	1	-0.7	0.1	153
1	1	0	0	0	0.4	-0.2	24
3	0	0.5	1	0	-0.1	0.3	39

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Answer.

We have B $= \{4, 1, 3\}$, and we need to determine the range of λ such that

$$\bar{b} + \lambda A_B^{-1} e_3 \geq 0$$

A_B^{-1} is the last three columns of the tableau, so

$$A_B^{-1} = \begin{bmatrix} 1 & -0.7 & 0.1 \\ 0 & 0.4 & -0.2 \\ 0 & -0.1 & 0.3 \end{bmatrix}$$

It is straightforward to verify that we must have $-130 \leq \lambda \leq 120$ and therefore $b3 \in [50, 300]$.

Exercise 3: Continue

(3). Find the range of the expected profit on special risk insurance such that the current basis remains optimal.

B	0	50	0	0	140	80	35400
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Answer.

The optimal basis remains the same if and only if $r_N - \lambda A_N^T (a_b^{-1})^T e_1 \geq 0$

$$50 \geq 0$$
$$140 - 0.4\lambda \geq 0$$
$$80 + 0.2\lambda \geq 0$$

We can find $A_B^{-1} A_N$ from the final tableau, and it is therefore straightforward to verify that $-400 \leq \lambda \leq 350$, and therefore that $c_1 \in [150, 900]$.

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